rMIDAS

rMIDAS package

- MIDAS is a Python package for missing-data imputation with deep learning
  - autoencoder
- rMIDAS relies on Python to implement the MIDAS imputation algorithm.

```r
library(rMIDAS)
```

```
Loading required package: data.table

Loading required package: mltools

Loading required package: reticulate

rMIDAS is loaded. Please read https://github.com/MIDASverse/rMIDAS for additional help on hov
```

```
##
## rMIDAS: Multiple Imputation using Denoising Autoencoders
## Authors: Thomas Robinson and Ranjit Lall
## Please visit https://github.com/MIDASverse/rMIDAS for more information
##
```

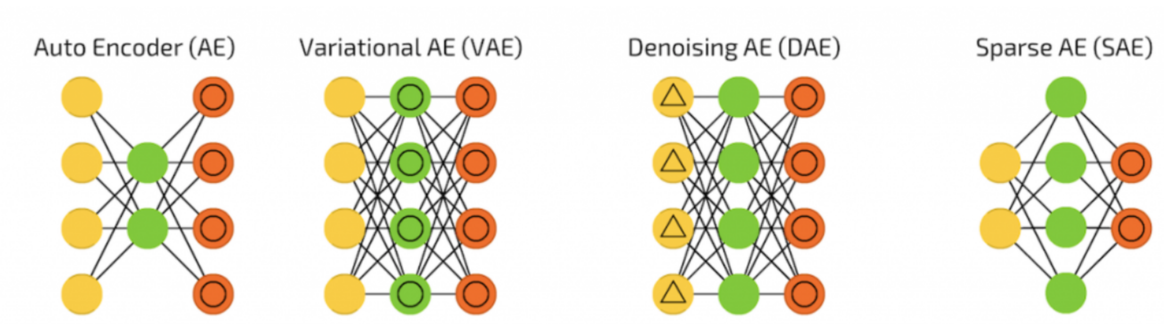
```r
library(finalfit)
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following object is masked from 'package:rMIDAS':

    combine

The following objects are masked from 'package:data.table':

    between, first, last

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

rMIDAS package

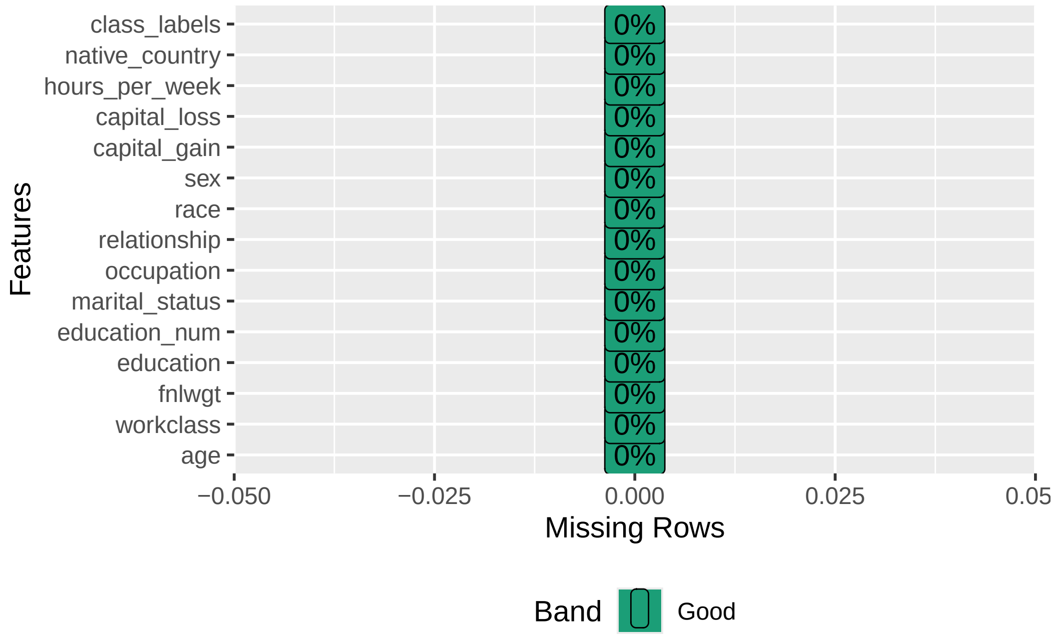
- MIDAS is a Python package for missing-data imputation with deep learning
  - autoencoder: denoising



Adult Census dataset

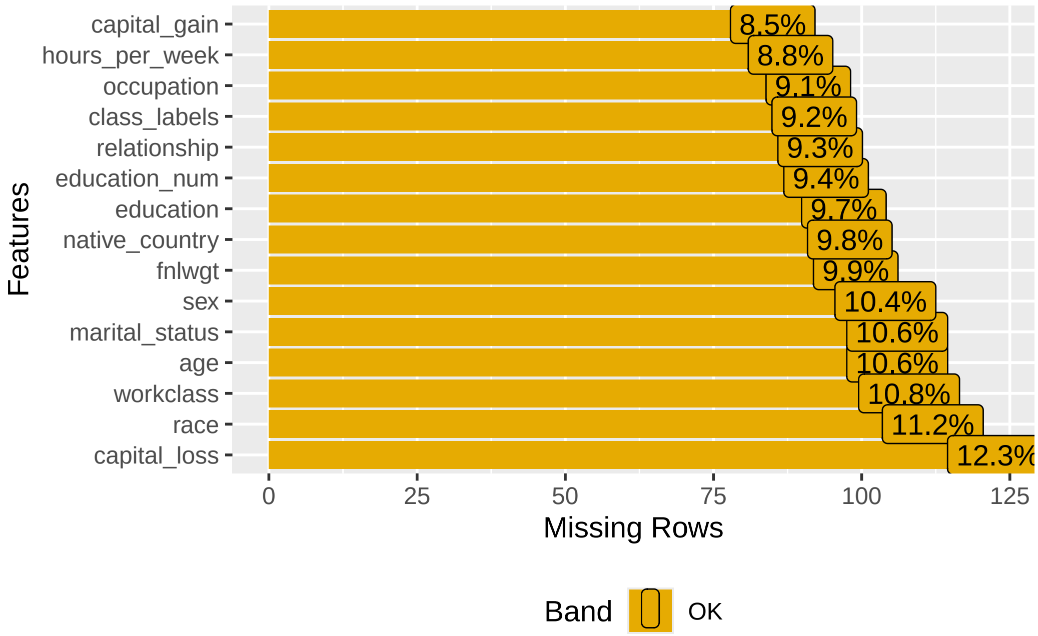
Show how to use the package to impute missing values in the Adult Census dataset

```
#adult <- read.csv("https://raw.githubusercontent.com/MIDASverse/MIDASpy/master/Examples/adul
adult <- read.csv("adult_data.csv", row.names = 1)[1:1000,]
DataExplorer::plot_missing(adult)
```



Add misssing values

```r
set.seed(89)
adult <- add_missingness(adult, prop = 0.1)
DataExplorer::plot_missing(adult)
```



Structure of data

```r
str(adult)
```

```
'data.frame':	1000 obs. of  15 variables:
 $ age           : int  39 NA 38 53 NA 37 49 52 NA 42 ...
 $ workclass     : chr  "State-gov" "Self-emp-not-inc" "Private" "Private" ...
 $ fnlwgt        : int  77516 83311 215646 234721 338409 NA 160187 209642 45781 159449 ...
 $ education     : chr  "Bachelors" NA "HS-grad" "11th" ...
 $ education_num : int  NA 13 9 7 13 14 5 9 14 13 ...
 $ marital_status: chr  "Never-married" "Married-civ-spouse" "Divorced" "Married-civ-spouse" 
 $ occupation    : chr  "Adm-clerical" "Exec-managerial" "Handlers-cleaners" "Handlers-clean
 $ relationship  : chr  "Not-in-family" "Husband" "Not-in-family" "Husband" ...
 $ race          : chr  "White" NA "White" "Black" ...
 $ sex           : chr  "Male" "Male" "Male" "Male" ...
 $ capital_gain  : int  2174 0 0 0 0 0 0 NA 14084 NA ...
 $ capital_loss  : int  0 0 0 NA 0 0 0 0 0 0 ...
 $ hours_per_week: int  40 13 40 40 40 40 16 45 50 40 ...
 $ native_country: chr  "United-States" "United-States" "United-States" NA ...
 $ class_labels  : chr  "<=50K" "<=50K" "<=50K" "<=50K" ...
```

Pre-processing

```
adult_cat <- c('workclass','marital_status',
               'relationship','race','education',
               'occupation','native_country')
adult_bin <- c('sex','class_labels')
adult_conv <- convert(adult,
                      bin_cols = adult_bin,
                      cat_cols = adult_cat,
                      minmax_scale = TRUE)
```

Train the model

```
adult_train <- train(adult_conv,
                       training_epochs = 20,
                       layer_structure = c(128,128),
                       input_drop = 0.75,
                       seed = 89)
```

```
Initialising Python connection

Size index: [6, 2, 7, 7, 6, 5, 16, 15, 28]

Computation graph constructed

Model initialised

Epoch: 0 , loss: 15.425114124051985
Epoch: 1 , loss: 11.575413134790235
Epoch: 2 , loss: 10.832636310208228
Epoch: 3 , loss: 10.304058890188895
Epoch: 4 , loss: 9.76571520682304
Epoch: 5 , loss: 9.31972320618168
Epoch: 6 , loss: 9.116158231612175
Epoch: 7 , loss: 8.958550560858942
Epoch: 8 , loss: 8.849149711670414
Epoch: 9 , loss: 8.70781163246401
Epoch: 10 , loss: 8.663574511005033
Epoch: 11 , loss: 8.485845996487525
```

```
Epoch: 12 , loss: 8.392313334249682
Epoch: 13 , loss: 8.251727434896655
Epoch: 14 , loss: 8.204197337550502
Epoch: 15 , loss: 8.09291801145
Epoch: 16 , loss: 8.006522778541811
Epoch: 17 , loss: 7.9091653285488
Epoch: 18 , loss: 7.851319905250303
Epoch: 19 , loss: 7.870906337614982
Training complete. Saving file...
Model saved in file: C:\Users\cili\AppData\Local\Temp\RtmpauTjya
```

Imputed datasets

- Generate 10 imputed datasets

```
adult_complete <- complete(adult_train, m = 10)
```

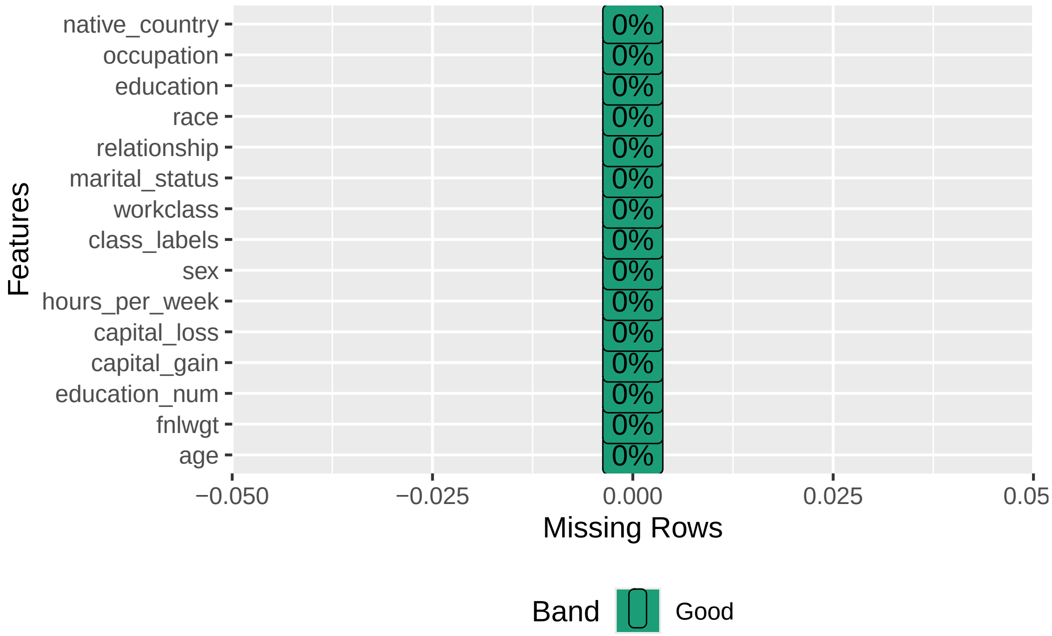
```
Model restored.
```

```
Imputations generated. Completing post-imputation transformations.
```

```
class(adult_complete)
```

```
[1] "list"
```

```
DataExplorer::plot_missing(adult_complete[[1]])
```



Works on the completed datasets

```r
adult_model <- rMIDAS::combine("class_labels ~ hours_per_week + sex",
                    adult_complete,
                    family = stats::binomial)
adult_model
```

```
            term    estimate   std.error statistic       df      p.value
1    (Intercept)  3.60381732 0.378015052  9.533529 153.8711 3.231071e-17
2 hours_per_week -0.04751552 0.008343815 -5.694700 135.4081 7.360395e-08
3        sexMale -0.73352239 0.186321476 -3.936864 778.4709 8.994744e-05
```